

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

March / April 2018

EC401 Data Communication & Networking

Date: 26.03.2018, Monday

Time: 10:00 a.m. to 1.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Use of scientific calculator is allowed.

SECTION-I

- Q-1 Answer Following Questions [05]**
- (a) What is protocol? Enlist three features of protocol. [02]
- (b) Explain client-server model involving request and reply. [03]
- Q-2 Attempt Any Three [15]**
- (a) What do you mean by network topology? Draw mesh, bus and star network topologies. [05]
- (b) Explain Guided media. [05]
- (c) Draw OSI Reference Model [05]
- (d) Enlist different error detection methods. [05]
- Q-3 Attempt Any Three [15]**
- (a) Define multiplexing. Discuss types of multiplexing. [05]
- (b) Compare LAN, MAN, WAN in detail. [05]
- (c) Explain TCP segment header in detail. [05]
- (d) Explain Sliding Window Protocol. [05]

SECTION – II

- Q-4 Attempt Following Questions [05]**
- (a) What do you understand by Throughput. Give example calculation of throughput. [02]
- (b) Explain simplex, half-duplex and full-duplex data flow of communication. [03]

- Q-5 Attempt Any Three [15]**
- (a) Explain firewall in detail [05]
 - (b) Define cryptography. Explain symmetric key and asymmetric key cryptography. [05]
 - (c) Describe RSA algorithm in cryptography. [05]
 - (d) Explain Digital signature with necessary illustration. [05]
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- Q-6 Attempt Any Three [15]**
- (a) Explain CSMA/CD with necessary flow chart. [05]
 - (b) What is the difference between static routing and dynamic routing algorithm? [05]
Explain Dynamic Routing.
 - (c) Explain High-level Data Link (HDLC) Protocol. [05]
 - (d) Mention need of DNS. Briefly explain DNS process. [05]